

CURRICULUM VITAE

Dale L. Forrister

Postdoctoral Fellow

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Educational Background

University of Utah

Ecology and Evolution

Ph.D., 2022

Lewis & Clark College

Biology

B.A., 2014

Research and Professional Appointments

2024-present	Stengl-Wyer Postdoctoral Scholar, U. of Texas at Austin
2022-2024	Postdoctoral Research Fellow, Smithsonian Tropical Research Institute
2021-2022	New Frontiers Graduate Fellow, U. of Illinois, National Center for Supercomputing Applications.
2016-2022	Ph.D. Candidate, U. of Utah, Salt Lake City, UT
2016-2017	Trainee, Global Change and Sustainability Fellow, University of Utah

Grants, Fellowships, and Awards

2024	<i>Stengl-Wyer Postdoctoral Fellowship</i> , U. of Texas at Austin (\$240,000)
2022	<i>George R. Riser Award for Outstanding Graduate Research</i> , U. of Utah
2021	<i>New Frontiers Graduate Fellowship</i> to use the Blue Waters petascale computing system, U. of Illinois, National Center for Supercomputing Applications (\$50,000)
2020	<i>National Geographic Society and Microsoft: Artificial Intelligence for Species Discovery</i> “The A,B,C's of species discovery through Artificial Intelligence, Botany and Chemistry” (Collaboration with Dr. Phyllis D. Coley, Dr. María José Endara and Dr. Juan Ernesto Guevara) (\$99,690)
2020	<i>Nexus Pilot Grant Sustaining Biodiversity: “Combine Machine Learning with Molecular Networking in Drug and Natural Product Discovery”</i> , U. of Utah (\$12,500)
2020	<i>Global Change and Sustainability Center Research Grant</i> , U. of Utah (\$3,000)
2019	<i>Thomas A. Kursar Memorial Fellowship</i> , U. of Utah (\$12,500)
2019	<i>Coley-Kursar Field Research Award</i> , U. of Utah (\$2,500)
2019	<i>NSF Graduate Research Internship Program (GRIP)</i> , Collaboration with the Smithsonian National Museum of Natural History (\$5,000)
2019	<i>CLAS Predissertation Field Research Grant</i> , U. of Utah (\$1,200)
2018	<i>Lewis and Clark Fund for Exploration and Field Research</i> , (\$5,000)
2018	<i>CLAS Predissertation Field Research Grant</i> , U. of Utah (\$2,000)
2018	<i>Global Change and Sustainability Center Research Grant</i> , U. of Utah (\$3,000)
2018	<i>Forest Global Earth Observatory Research Grants Program</i> , Smithsonian Institution (\$10,000)
2018	<i>Sustainable Campus Initiative Fund Grant: “Phenology Working Group”</i> , U. of Utah (\$10,000)
2018	<i>George R. Riser Travel Award</i> , U. of Utah (\$800)
2017	<i>Global Change and Sustainability Center Research Grant</i> , U. of Utah (\$3,000)
2017	<i>National Geographic Young Explorer Grant</i> , Collaboration with Adrian Curbelo Diaz “Documenting the importance of honeybee production in Camaguey, Cuba” (\$5,000)
2017	<i>NSF Graduate Research Opportunities Worldwide (GROW)</i> , Collaboration with e-phenology São Paulo State University, Brazil (\$5,000)

2016 NSF Graduate Research Fellowship (GRFP) (\$138,000)
 2016 Global Change and Sustainability Center Fellowship, U. of Utah (\$27,000)

Publications [ORCID iD: 0000-0001-8170-7187](#)

17. Cárdenas, R.E., Rodríguez-Ortega, C., Utreras, D., **Forrister, D.L.**, Endara M.-J., Queenborough, S.A., Alvia, P., Mendéndez-Guerrero, P.A., Báez, S., Donoso, D.A. (2024) Long-term strict ant-plant mutualism identity characterizes growth rate and leaf shearing resistance of an Amazonian myrmecophyte. *Scientific Reports*. 14 (1), 17813
16. Endara, M.-J., **Forrister, DL**, Coley, PD, (2023). The ecolutionary ecology of plant chemical defenses: from molecules to communities. *Annual Review of Ecology, Evolution and Systematics*. 107-127 <https://doi.org/10.1146/annurev-ecolsys-102221-045254>
15. Walker, T. W. N., Schrodtt, F., Allard, P. M., Defosse, E., Jassey, V. E. J., Schuman, M. C., Alexander, J. M., Baines, O., Baldy, V., Bardgett, R. D., Capdevila, P., Coley, P. D., van Dam, N. M., David, B., Descombes, P., Endara, M. J., Fernandez, C., **Forrister, DL.**, Gargallo-Garriga, A., ... Peñuelas, J. (2023). Leaf metabolic traits reveal hidden dimensions of plant form and function. *Science Advances*, 9(35), 1–14. <https://doi.org/10.1126/sciadv.adi4029>
14. Rice, M. C., Little, J. H., **Forrister, D. L.**, Machado, J., Clark, N. L., & Gagnon, J. A. (2023). Gadusol is a maternally provided sunscreen that protects fish embryos from DNA damage. *Current Biology*, 33(15), 3229-3237.e4. <https://doi.org/10.1016/j.cub.2023.06.012>
13. **Forrister, D.L.**, Endara, M., Soule, A.J., Younkin, G.C., Mills, A.G., Lokvam, J., et al. (2022). Diversity and Divergence: Evolution of secondary metabolism in the tropical tree genus *Inga* . *New Phytol.*, 631–642.
12. Tozetto, L., **Forrister, D.L.**, Duval, M., Hays, T., Garwood, N.C., Vargas, R., et al. (2022). Army ant males lose seasonality at a site on the equator. *Biotropica*.
11. Guevara Andino, J.E., Hernández, C., Valencia, R., **Forrister, D.** & Endara, M.-J. (2022). Accelerating the discovery of rare tree species in Amazonian forests: integrating long monitoring tree plot data with metabolomics and phylogenetics for the description of a new species in the hyperdiverse genus *Inga* Mill. *PeerJ*, 10, e13767.
10. Donoso, D.A., Basset, Y., Shik, J.Z., **Forrister, D.L.**, Uquillas, A., Salazar-Méndez, Y., et al. (2022). Male ant reproductive investment in a seasonal wet tropical forest: Consequences of future climate change. *PLoS One*, 17, 1–13.
9. Bradshaw, A.J., Dentinger, B., Backman, T., Ramírez-Cruz, V., **Forrister, D.**, Winter, J., et al. (2022). DNA Authentication and Chemical Analysis of *Psilocybe* Mushrooms Reveal Widespread Taxonomic Misdeterminations and Inconsistencies in Metabolites. *Appl. Environ. Microbiol.*, 88, e01498-22.
8. Endara, M.-J., **Forrister, D.**, Nicholls, J., Stone, G.N., Kursar, T. & Coley, P. (2022). Impacts of Plant Defenses on Host Choice by Lepidoptera in Neotropical Rainforests. In: *Caterpillars in the Middle: Tritrophic Interactions in a Changing World* (eds. Marquis, R.J. & Koptur, S.). Springer International Publishing, Cham, pp. 93–114.

7. Endara, M.J., Soule, A.J., **Forrister, D.L.**, Dexter, K.G., Pennington, R.T., Nicholls, J.A., et al. (2022). The role of plant secondary metabolites in shaping regional and local plant community assembly. *J. Ecol.*, 110, 34–45.
6. Walker T, Alexander JM, Allard P-M, Baines O, BALDY V, Bardgett R, Capdevila P, Coley PD, David B, Defosse E, Ernst M, Fernandez C, **Forrister DL**, Gargallo-Garriga A, Jassey V, Marr S, Neumann S, Pellissier L, Penuelas J, Peters K, Rasmann S, Roessner U, Sardans Galobard J, Schrod F, Schuman M, Soule A. (2022). Functional Traits 2.0: The power of the metabolome for ecology. *J. Ecol.*, 110, 4–20.
5. **Forrister, D.L.**, Endara, M.J., Younkin, G.C., Coley, P.D. & Kursar, T.A. (2019). Herbivores as drivers of negative density dependence in tropical forest saplings. *Science* (80-.), 363, 1213–1216.
4. Schneider, G.F., Coley, P.D., Younkin, G.C., **Forrister, D.L.**, Mills, A.G. & Kursar, T.A. (2019). Phenolics lie at the centre of functional versatility in the responses of two phytochemically diverse tropical trees to canopy thinning. *J. Exp. Bot.*, 70, 5853–5864.
3. Endara, M.J., Nicholls, J.A., Coley, P.D., **Forrister, D.L.**, Younkin, G.C., Dexter, K.G., et al. (2018). Tracking of host defenses and phylogeny during the radiation of neotropical Inga-feeding sawflies (Hymenoptera; Argidae). *Front. Plant Sci.*, 9.
2. Endara, M.J., Coley, P.D., Wiggins, N.L., **Forrister, D.L.**, Younkin, G.C., Nicholls, J.A., et al. (2018). Chemocoding as an identification tool where morphological- and DNA-based methods fall short: Inga as a case study. *New Phytol.*, 218, 847–858.
1. Wiggins, N.L., **Forrister, D.L.**, Endara, M.J., Coley, P.D. & Kursar, T.A. (2016). Quantitative and qualitative shifts in defensive metabolites define chemical defense investment during leaf development in Inga, a genus of tropical trees. *Ecol. Evol.*, 6, 478–492.

Research Experience

Postdoctoral Fellow

- 2023-24 Ecometabolomics, environmental gradients and the growth-defense trade-off in tropical trees. Smithsonian Tropical Research Institute and University of Texas at Austin

PhD Candidate

- 2021 Using the Blue Waters Petascale Computer system to process Drone Imagery and track new leaf production in an ever-wet tropical rainforest.
- 2017 -22 Does plant phenology drive insect seasonality in an aseasonal tropical forest? Ph.D. Dissertation Fieldwork in Yasuní, Ecuador, University of Utah, Advisors: Dr. Phyllis D. Coley and Dr Thomas A. Kursar
- 2021 Estimating chemical diversity across the Ecuadorian Amazon using a forest plot network and untargeted metabolomics. University of Utah, Universidad Tecnológica Indoamerica and Universidad de las Americas. Advisor Dr. Phyllis D Coley, Collators: Dr. María José Endara and Dr. Juan Ernesto Guevara
- 2020 Combining machine learning and molecular networking to improve annotation of unknown compounds in metabolomic experiments. University of Utah, Advisor Dr. Phyllis D Coley, Collaborator Dr. Bei Wang
- 2015 -20 Characterize plant secondary metabolite profile using untargeted metabolomics. University of Utah, PI: Dr. Phyllis D Coley and Dr Thomas A. Kursar

Undergraduate

- 2014 Evolution of defense traits and the study of plant herbivore-interactions in Yasuní, Ecuador, University of Utah, PI: Dr. Phyllis D Coley and Dr Thomas A. Kursar
- 2014 Study the generation of a Biodiversity Hotspot through the biogeographic analysis of Caribbean arachnid lineages. Lewis & Clark College, PI. Greta J. Binford
- 2013 Field assistant: Collected arachnid specimens in Choco, Colombia using a variety of collection methods for use in phylogenetic analysis of Caribbean Biogeography. Lewis & Clark College PI. Dr. Greta J. Binford and Dr. Lauren A. Esposito

Botanical Collections Management and Fieldwork

- Managed a team of botanists conducting a nation-wide botanical survey of Panamanian Flora for (eco)metabolomics, transcriptomics and genomics. Spearheaded the curation and database architect for a collection of > 5,000 botanical vouchers, and associated tissue archives deposited in the Panamá's National Herbarium at the University of Panama. Project includes the establishment of an ultra-cold biobank for each voucher enabling transcriptomics and genomic analysis. (2022-2024)
- PI, on multi-year (2018-2021) campaign characterizing the spatial-temporal dynamics plant-herbivore interactions between lepidoptera and the tropical tree genus *Inga* (Fabaceae). Managed collections, permitting and export and barcoding of 4,000 tropical lepidoptera.
- Experienced tropical ecologist and field botanist: Costa Rica (2017), Ecuador (2018-2021), Panama 2022-2024)

Seminars and Conference Presentations

Invited Symposia and Colloquia

- 2024 International Botanical Conference, Madrid, Spain. Oral presenter as part of the colloquia titled: *Diversification in large dominant plant lineages: integration of phylogeny, ecology, physiology, and biogeography*.
- 2024 Barro Colorado Island Centennial Symposium, Gamboa, Panama. Oral presenter as part of the colloquia titled: *The role of plant-enemy interactions in shaping tropical forest communities across environmental gradients*.
- 2023 Plant-Herbivore Interaction Gordon Research Seminar, Ventura, California. Invited oral presentation.
- 2022 Forrister DL. "Exploring the phytochemical landscape in space and time: implications for the evolution of tropical trees and species coexistence" Association for Tropical Biology and Conservation. Cartagena, Colombia. Oral presenter as part of the colloquia titled: *The role of biotic interactions in shaping tropical forest diversity*.
- 2022 Evolution

Invited Seminars

- 2023 Forrister DL. Diversity and divergence: evolution of secondary metabolism in the tropical tree genus *Inga*, Tupper Seminar. Smithsonian Tropical Research Institute, Panama.
- 2022 Forrister DL. Leafing Phenology and Insect Seasonality in an Ever-wet Tropical Forest. New Frontiers Initiative Webinar at the University of Illinois Urbana-Champaign. (<https://newfrontiers.illinois.edu/leafing-phenology-and-insect-seasonality-in-an-ever-wet-tropical-forest/>)
- 2020 *Forrister DL. "Plant-Herbivore Interactions: Plant defense and Tropical Tree Diversity". Conservatorio de biodiversidad. Universidad Tecnológica Indoamérica. Quito, Ecuador. Virtual presentation.

* Spanish language seminar

Selected Conference Talks and Posters

- 2020 Forrister DL, Endara MJ, Younkin GC, Coley PD and Kursar TA. “Herbivores as drivers of negative density dependence in tropical forest Saplings”. Oral Presentation
- 2019 Endara MJ, Forrister DL, Younkin GC, Nicholls JA, Stone G, Soule AJ, Dexter KG, Pennington RT, Lemes M, Kidner CA, Kursar TA, Coley PD. “Reciprocal impacts of plant defenses and host choice by Lepidoptera in Neotropical rainforests”. Entomological Society of America, St. Louis, Missouri. Oral Presentation.
- 2019 Forrister DL, Endara M-J, Coley PD, Wiggins NL, Younkin GC, Nicholls JA, Dexter KG, Kidner CA, Stone GN, and Kursar TA. “Chemocoding as an identification tool where morphological and DNA-based methods fall short: Inga as a case study”. International Biogeography Society, Quito Ecuador. Poster.
- 2017 Forrister DL. “Does resource competition or pest pressure drive negative density dependence? Evidence from the tree genus Inga (Fabaceae) on Barro Colorado Island, Panama” Biology Department Retreat, Univ. of Utah, Salt Lake City, Utah. 5-Minute Lightning Talk
- 2017 Forrister DL, Endara MJ, Kursar TA and Coley PD. “Does resource competition or pest pressure drive negative density dependence? Evidence from the tree genus Inga (Fabaceae) on Barro Colorado Island, Panama” doi: [10.7490/f1000research.1114833.1](https://doi.org/10.7490/f1000research.1114833.1). Ecological Society of America, Portland, Oregon, Poster
- 2017 Forrister DL. “Does resource competition or pest pressure drive negative density dependence? Evidence from the tree genus Inga (Fabaceae) on Barro Colorado Island, Panama” Gordon Research Conference: Plant-Herbivore Interactions, Ventura, California. Poster
- 2016 Forrister DL. “Chemocoding as an identification tool where morphological and DNA-based methods fall short: Inga as a case study”. American Society of Mass Spectrometry, San Antonio, Texas. Poster

Teaching Experience

- 2024 Smithsonian Tropical Research Institute – Experimental Design Workshop, Rampup Fellows Program.
- 2020-24 Ecometabolomics Course, creator and lead instructor of three-day introductory course in metabolomics. (https://github.com/dlforrister/ICIPE_Metabolomics_WS_2022)
- *2024 - Gamboa, Panama - Smithsonian Tropical Research Institute
- *2022 - Nairobi, Kenya - International Center for Insect Physiology and Ecology
- *2020 - Quito, Ecuador - Universidad de Las Americas.
- *2021 Instructor: Conservation Biology in Latin America - Culture and Language Across Curriculum (CLAC) section for Conservation Biology (BIOL3470).
- 2021 Teaching Assistant Conservation Biology (BIOL 3470)
- 2019 Teaching Assistant Field Botany (BIOL 2355)
- 2018 Teaching Assistant Field Botany (BIOL 2355)
- * Spanish language course

Outreach and Service

Public Lectures

- 2020 Stem Ambassador Program Fellow: Received training on public engagement and facilitated an engagement activity about plant compounds and homebrewing.
- 2020 Initiative to bring Science Programs to the Incarcerated (Inspire) lecture at the Draper State Prison and Salt Lake County Jail.

- 2019 Radio Interview: Undisciplined: The Tropical Ecologist and the Microbiologist (www.upr.org/post/undisciplined-tropical-ecologist-and-microbiologist)
- 2018/19 Invited Speaker: Pinhead Institute's Scholars in the Schools program Telluride, Colorado Visited 6th - 9th grade science classes at Norwood High School and Paradox Valley Charter School to lead laboratory activity focused on investigating the many ways in which plants defend themselves against herbivore and pathogen attack.
- 2018 Guest Lecturer: Insect seasonality in an ever-wet tropical rain forest, Yasuní Research Station, Yasuní, Ecuador.
- 2017 Guest Lecturer: Plant-Herbivore Interactions in Tropical Rain Forests, São Paulo State University, environmental science field course, Serra do Japi, Jundiá, São Paulo Brazil.
- 2016 Guest Lecturer: Plant-Herbivore Interactions in Tropical Rain Forests, AP Environmental Science at Highland High School taught by Monica French
- 2016 Granite District Senior Shadowing Day, Univ. of Utah
- 2012 Environmental Educator, Corporación Laguna Verde. Valparaíso, Chile

Service

- 2016-22 Manuscript reviewer for Journal of Ecology, Oecologia, Journal of Vegetation Science, Proceedings of the Royal Society B, ELife, PeerJ
- 2021-22 College of Science wide GSAC committee on health insurance
- 2020-21 Co-chair Biology Graduate Student Advisory Committee (GSAC)
- 2020-21 Organizer School of Biological Sciences Annual Retreat
- 2020-21 Chair of Retention, Promotion and Tenure (RPT) Graduate Student Committee
- 2016-17 Co-chair of Scientific Speaking weekly seminar series, Univ. of Utah

Professional Trainings

- 2018/20 Wilderness First Aid Certification
- 2017 Organization for Tropical Studies Course: Sistemática de Plantas Tropicales Five-week intensive graduate-level course organized by OTS and University of Costa Rica focused on tropical plant systematics and field identification. The course was taught in Spanish and included participants from 9 Latin American countries and the United States.

Professional Societies

- 500 Queer Scientists (<https://500queerscientists.com/>)
- Association for Tropical Biology and Conservation
- Ecological Society of America

Faculty Mentors

Postdoctoral advisors

- S. Joseph Wright, Senior Staff Scientist, Smithsonian Tropical Research Institute
- Brian E. Sedio, Assistant Professor, U. of Texas at Austin
- Annette Osling, Associate Professor, U. of Texas at Austin

PhD advisor

- Phyllis (Lissy) D. Coley, Distinguished Professor, U. of Utah
- Thomas A. Kursar, Professor, U. of Utah

PhD committee members

- Donald Feener, Professor, U. of Utah
- Bryn Dentinger, Associate Professor, U. of Utah
- Phil Dennison, Professor, U. of Utah
- Fred Adler, Professor, U. of Utah